

$$= \begin{bmatrix} w'_{11}x_{i1} + w'_{21}x_{i2} + w'_{31}x_{i3} + w'_{41}x_{i4} \\ w'_{12}x_{i1} + w'_{22}x_{i2} + w'_{32}x_{i3} + w'_{42}x_{i4} \\ w'_{13}x_{i1} + w'_{23}x_{i2} + w'_{33}x_{i3} + w'_{43}x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$\Rightarrow \sigma \left(\begin{bmatrix} w'_{11}x_{i1} + w'_{21}x_{i2} + w'_{31}x_{i3} + w'_{41}x_{i4} + b_{11} \\ w'_{12}x_{i1} + w'_{22}x_{i2} + w'_{32}x_{i3} + w'_{42}x_{i4} + b_{12} \\ w'_{13}x_{i1} + w'_{23}x_{i2} + w'_{33}x_{i3} + w'_{43}x_{i4} + b_{13} \end{bmatrix} \right)$$

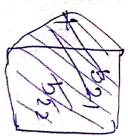
$$= \begin{bmatrix} 0_{11} \\ 0_{12} \\ 0_{13} \end{bmatrix} \rightarrow \text{output of layer 1.}$$

$$= a^{[1]} \rightarrow \text{activation of layer 1.}$$

layer 2.

$$\underbrace{\begin{bmatrix} w''_{11} & w''_{12} \\ w''_{21} & w''_{22} \\ w''_{31} & w''_{32} \end{bmatrix}}_{(3,2)} \cdot \underbrace{\begin{bmatrix} 0_{11} \\ 0_{12} \\ 0_{13} \end{bmatrix}}_{(2,1)} + \underbrace{\begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}}_{(2,1)}$$

$$= \begin{bmatrix} w''_{11}0_{11} + w''_{21}0_{12} + w''_{31}0_{13} + b_{21} \\ w''_{12}0_{11} + w''_{22}0_{12} + w''_{32}0_{13} + b_{22} \end{bmatrix}$$



$$\Rightarrow \sigma \left(\begin{bmatrix} w''_{11}0_{11} + w''_{21}0_{12} + w''_{31}0_{13} + b_{21} \\ w''_{12}0_{11} + w''_{22}0_{12} + w''_{32}0_{13} + b_{22} \end{bmatrix} \right) = \begin{bmatrix} 0_{21} \\ 0_{22} \end{bmatrix}$$

$\rightarrow$ Output of layer 2

$= a^{[2]} \rightarrow$ activation of layer 2

layer 3

$$\begin{bmatrix} w'''_{11} \\ w'''_{12} \end{bmatrix}^T \cdot \begin{bmatrix} 0_{21} \\ 0_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix}$$

$(2,1) \cdot (1,2) \rightarrow (2,1)$

$$\Rightarrow \sigma \left(\begin{bmatrix} w'''_{11}0_{21} + w'''_{12}0_{22} + b_{31} \end{bmatrix} \right) = 0_{31} = \hat{y}_1 \rightarrow \text{final output}$$

$= a^{[3]} \rightarrow \text{activation 3.}$

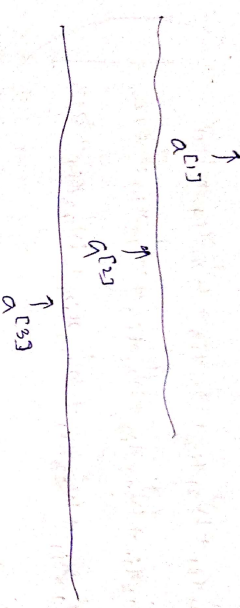
$$a^{[0]} = x^{[0]}$$

$$a^{[1]} = \sigma(a^{[0]}w^{[0]} + b^{[1]})$$

$$a^{[2]} = \sigma(a^{[1]}w^{[1]} + b^{[2]})$$

$$a^{[3]} = \sigma(a^{[2]}w^{[2]} + b^{[3]})$$

$$\sigma \left(\sigma(a^{[0]}w^{[0]} + b^{[1]})w^{[1]} + b^{[2]} \right)w^{[2]} + b^{[3]}$$



18/07/25

What is Loss fn?

Loss fn is a method of evaluating how well your algorithm is modelling your dataset.

L (parameters) $\rightarrow$ high $\rightarrow$ poor, small $\rightarrow$ great

$f(x) = x_1x_2$

$$L = (y_i - \hat{y}_i)^2$$

$\uparrow$

$mx_1 + b$

$L(m,b) = \text{min.}$

Why is Loss fn important?

[You can't improve what you can't measure]

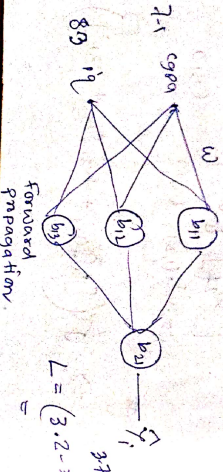
Peter Dinkler

Loss function $\rightarrow$ eye of ML algorithm

Why fn in Deep learning?

copy iq message (100)

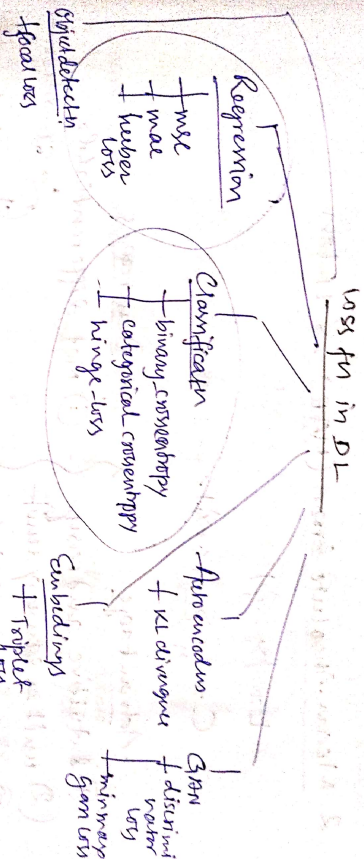
71	8.5
8.5	91
6.9	102
5.1	89



Forward Propagation $\rightarrow \hat{y}_i \rightarrow$ Loss $\rightarrow$ determine new value of w & b using Gradient Descent

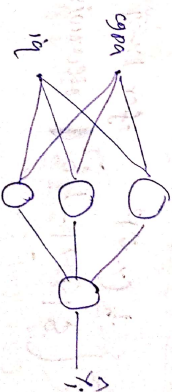
GD: $w \leftarrow w - \eta \cdot \frac{\partial L}{\partial w}$ $b \leftarrow b - \eta \cdot \frac{\partial L}{\partial b}$ $\leftarrow$ Forward prop. $\leftarrow$ new w, b

if epochs = 10
8 no of rows = 5
total 10x5 = 50 updates in w & b



Loss fn vs Cost fn

cgpa	iq	percentage	predn
6.3	100	6.3	6.1
7.1	91	4.1	4
8.5	83	3.5	3.4
9.2	102	7.2	7



Loss fn $\rightarrow$ single training ex.
Error $y = 6.3, \hat{y}_1 = 6.1$
 $(y_1 - \hat{y}_1)^2$

Cost fn $\rightarrow$ for whole data.
 $\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
 $= \frac{(6.3 - 6.1)^2}{4}$

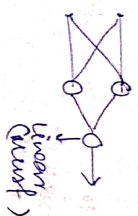
Mean squared error (MSE)

Squared loss, L^2 loss.

cgpa	iq	percentage	y_i	$\hat{y}_i$
6.3	100	6.3	6.3	6.1
7.1	91	4.1	4.1	4
8.5	83	3.5	3.5	3.4
9.2	102	7.2	7.2	7



- Advantage
- Easy to interpret
 - Differentiable
 - Local minima
- Disadvantage
- Error unit $\rightarrow$ squared
 - Robust to outliers



2. Mean Absolute Error (MAE) $\rightarrow$ L1 loss

$$L = \sum |y_i - \hat{y}_i|$$

$$C = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Advantage

- Intuitive and easy
- unit $\rightarrow$ same y unit
- Robust to outliers

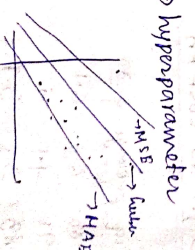
Disadvantage

- Not differentiable

GD $\rightarrow$ differentiation.
Sub Gradient

3. Huber Loss

$$L = \begin{cases} \frac{1}{2} (y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta \\ \delta |y - \hat{y}| - \frac{1}{2} \delta^2 & \text{otherwise} \end{cases}$$



4. Binary Cross Entropy (log-loss)

$\rightarrow$ Classification
 $\rightarrow$ 2 classes

$$\text{Loss fn} = -\gamma \log(\hat{\gamma}) - (1-\gamma) \log(1-\hat{\gamma})$$

$\gamma \rightarrow$ actual
 $\hat{\gamma} \rightarrow$ prediction



$$\text{Cost fn} = -\frac{1}{n} \sum_{i=1}^n \gamma_i \log(\hat{\gamma}_i) + (1-\gamma_i) \log(1-\hat{\gamma}_i)$$

Advantage

- Differentiable

Disadvantage

- Multiple local minima
- Not intuitive

5. Categorical cross entropy [used in softmax regression]

→ [Multi-classes] {classification}

$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

(K = no. of classes)

qypr	iq	placed	Yes	No	Maybe
8	80	Yes	1	0	0
6	60	No	0	1	0
7	70	Maybe	0	0	1

$$L = - \gamma_1 \log(\hat{y}_1) - \gamma_2 \log(\hat{y}_2) - \gamma_3 \log(\hat{y}_3)$$

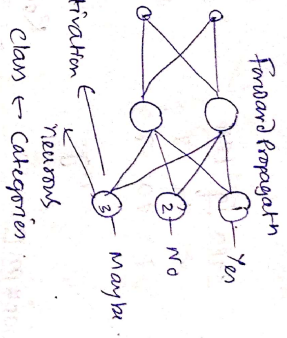
for 1st

$$Z_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

Activation ← neurons

for no →

$$\frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}} \text{ for Maybe} \rightarrow \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$



Sparse categorical cross entropy

OHE

qypr	iq	placed	Yes	No	Maybe
7	70	Yes	1	0	0
8	80	No	0	1	0
6	60	Maybe	0	0	1

Integer encode:

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ because } (y_1 \neq y_2 = y_3 = 1)$$

$$\textcircled{1} \rightarrow \begin{bmatrix} 7 \\ 30 \end{bmatrix} \rightarrow -1 \times \log(0.1)$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 8 \\ 80 \end{bmatrix} \rightarrow -1 \times \log(0.4)$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 6 \end{bmatrix} \rightarrow \begin{bmatrix} 3 \\ 2 \end{bmatrix} \rightarrow \begin{bmatrix} 6 \\ 60 \end{bmatrix} \rightarrow -1 \times \log(0.6)$$

$$L \rightarrow (6, 60) \rightarrow -\gamma_3 \log(\hat{y}_3)$$

$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

$$C = - \frac{1}{\gamma} \sum_{j=1}^K y_j \log(\hat{y}_j)$$

Summary

Reg → MSE

→ Activations → MAB

→ hidden layer

Classification

→ binary → binary cross entropy (BCE)

→ few cat → categorical cross entropy (CCE)

→ many " → (SCE) → sparse

Back propagation

Prerequisite

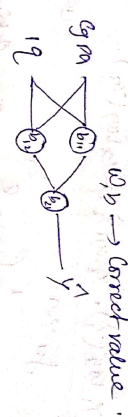
→ Gradient descent

→ forward propagation

Backprop
↓
Apply → train NN

qypr	iq	placed
8	80	8
7	70	7

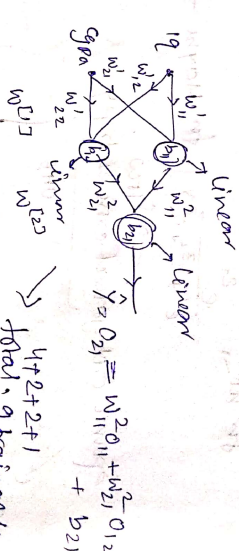
reg.



backprop.

Regression

iq	qypr	placed
80	8	8
60	9	5
70	5	8
120	7	11



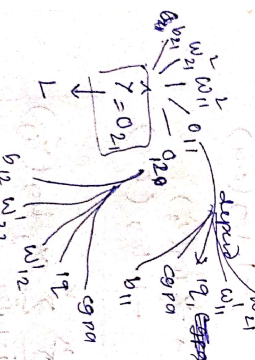
Steps: 0) init w, b (let w=1, b=0)

1) you select a pt. (zero)

2) Predict(LPP) → forward propagation

3) choose a loss fn. → MSE

4) w, b - update



$$w_{old} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$$

derivative of loss

$$w_{11}^2 = w_{11}^2 - \eta \frac{\partial L}{\partial w_{11}^2}$$

$$w_{21}^2 = w_{21}^2 - \eta \frac{\partial L}{\partial w_{21}^2}$$

$$b_{21}^{new} = b_{21}^{old} - \eta \frac{\partial L}{\partial b_{21}^{old}}$$

→ trainable parameters